

Hugo HADJUR

20-12-1995

PhD in Informatics

Thesis defended on July 13th, 2023

CONTACT

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Thesis manuscript



Academic Page



Google Scholar



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PUBLICATIONS

Hugo Hadjur, Doreid Ammar, Laurent Lefèvre. **Services Orchestration at the Edge and in the Cloud for Energy-Aware Precision Beekeeping Systems**, 2023 IEEE International Parallel and Distributed Processing Symposium Workshops (IPDPSW), St. Petersburg, FL, USA, 2023, pp. 769-776, [10.1109/IPDPSW59300.2023.00129](https://doi.org/10.1109/IPDPSW59300.2023.00129). [hal-04091575](https://hal.archives-ouvertes.fr/hal-04091575)

Hugo Hadjur, Doreid Ammar, Laurent Lefèvre. **Toward an intelligent and efficient beehive: A survey of precision beekeeping systems and services**, Computers and Electronics in Agriculture, Elsevier, 2022, 192, pp. 1-16. [10.1016/j.compag.2021.106604](https://doi.org/10.1016/j.compag.2021.106604). [hal-03483914v2](https://hal.archives-ouvertes.fr/hal-03483914v2)

Kazuma Mori, **Hugo Hadjur**, Masahiko Haruno. **Natural Language Content Mediates the Association Between Active Interactions on Social Network Services and Subjective Well-Being**, Cyberpsychology, Behavior, and Social Networking, 2022, pp. 678-685. [10.1089/cyber.2021.0340](https://doi.org/10.1089/cyber.2021.0340)

Hugo Hadjur, Doreid Ammar, Laurent Lefèvre. **Analysis of Energy Consumption in a Precision Beekeeping System**. IoT '20 - 10th International Conference on the Internet of Things, Oct 2020, Malmö, Sweden. pp. 1-11, [10.1145/3410992.3411010](https://doi.org/10.1145/3410992.3411010). [hal-02973772](https://hal.archives-ouvertes.fr/hal-02973772)

WORK EXPERIENCE

SEPT. 2020 - AUG. 2023
3-year CDD



PhD Candidate and Assistant Professor at **École normale supérieure de Lyon** & **aivancity** (FRANCE)

— PhD Research Topic: **Designing sustainable, autonomous, and energy efficient Internet of Things systems**, applied to smart beehives

PhD under the advisorship of **Laurent Lefèvre** and **Doreid Ammar** (LIP-Avalon laboratory)

— Teaching courses: **advanced data analysis and visualization**, **data preprocessing and visualization** and **coding for AI and data science** (Python, Tableau)

Head of online course platform **aivancityX**

SEP. 2019 - AUG. 2020
1-year CDD



Instructor & Researcher at **emlyon Business School** (FRANCE)

— Research: Connected beehives, bees' data analysis (**IoT**, **machine learning**, **computer vision**)

— Teaching: **Python programming bootcamp** and **business analyst toolbelt** (SQL, Excel, Tableau)

JAN. 2019 - SEP. 2019
8-month VIE



Data Analyst at **Schneider Electric**, Houston TX (USA)

— Supported the business by building **reports/dashboards and KPIs** to monitor performance

— Conducted **database integration** to form practical recommendations

— **Communicated findings** and insights to cross-functional team members and management

— Tool used: **SQL**, **Tableau**, **Python**

OCT. 2018 - DEC. 2018
3-month CDD



Part-time Instructor at **emlyon Business School** (FRANCE)

— Teaching: **Python programming bootcamp**

OCT. 2017 - SEP. 2018
1-year exchange



Student Researcher at **Kyoto University's Ishii Lab** (JAPAN)

— Master's thesis: master board games thanks to **Reinforcement Learning** (Q-Learning, SARSA, Deep Reinforcement Learning)

MAY-SEP. 2017
4-month internship



Data Science Research Internship at **CiNet** (Center for Information and Neural Networks, Osaka, JAPAN)

— **Conducted research** about the links between **Twitter data**, human behavior and brain data

— **Analyzed large data sets** ($\approx 500,000$ Twitter users) thanks to **PCA, clustering & regression** techniques

— Worked in a **Japanese environment**

PROGRAMMING & SOFTWARE

PROGRAMMING:

Expert knowledge: **Python** (data analysis, visualization and deep learning: time series, computer vision and reinforcement learning), **SQL**, **Tableau**, **R**

Advanced: **Javascript** (full-stack), **Java**, **C**, **C++**, **MATLAB**

Experience using: **Rest APIs**, **Git**

DESIGN & TEXT:

L^AT_EX, **Adobe Photoshop**, **Microsoft Office**, **Inkscape**

QUALIFICATIONS

✓ “GRANDE ÉCOLE” DIPLOMA + PHD

2020-2023



PhD at **ENS de Lyon** (**LIP-Avalon** laboratory) & **aivancity**

The goal of this thesis is to develop and model Internet of Things (IoT) systems and services based on Artificial Intelligence (AI) to support beekeepers in their work and preserve bees while studying the efficiency of connected beehives that are autonomous, sustainable, distributed and operate under limited energy budget.

2017-2018



Exchange year at **Kyoto University's Graduate School of Informatics**

Alongside courses in English and Japanese, I worked toward my Master's thesis. This thesis proposes a reinforcement learning method applied to the game of bridge. The Q-Learning and SARSA techniques reach a level of play comparable to an intermediate human player.

— Classes in **English & Japanese** in the department of **social informatics: statistics, machine learning**

2015-2017



French “Grandes Ecoles”: **Ensimag, Grenoble Institute of Technology** (equivalent: Master's Degree)

— Solid and rigorous skill set in **programming, advanced statistics, data mining, database manipulation** among others

2013-2015



Scientific preparatory classes at **CPP Grenoble** (equivalent: Bachelor's Degree)

— 2-year intensive university-level preparatory course to enter **highly competitive Ivy League engineering schools**

2007-2013



Middle school and high school in Tokyo (4 years at **LFIT**) and in Singapore (1 year at **LFS**)

— Scientific baccalaureate with highest honours (French school)